Homework 8

Bode Plots
Do the following book problems:

- **6.3a,d** only. Label the slopes on the magnitude vs. ω plot. Label the phase at $\omega = 0$ and $\omega = \infty$. Use Matlab to aid you if needed, but you should be able to determine the slopes and phases intuitively as done in class.
- **6.7e** only via Matlab. Calculate the phase at $\omega = \infty$ analytically, and confirm the Matlab bode plot shows the same limit.

Nyquist Stability

- **6.20**
- **6.21**. You can use Matlab to make this easy. Consider negative k.
- **6.23d,g** only. Make Nyquist plots of the system, and determine ranges of k where the closed loop system will be stable. You can use Matlab to make this easy. Consider negative k. For part g), use a root locus plot to confirm your answer, and label the critical gains where the poles cross the imaginary axis.

Optional Bonus
Create an open loop system $kG(s)$ with at least 3 poles that, when put in a unity feedback system, has a finite range of stability in terms of the system gain k: $c_1 < k < c_2$ for stability. Make a Nyquist plot and root locus plot and explain the system behavior in each region of k.